Azure Databricks – Delta Table Hands-on Exercises

Scenario: Fitness Club Memberships

Step 1 – Create the Delta table

Run this in an Azure Databricks notebook SQL cell.

```
%sql
CREATE OR REPLACE TABLE fitness_memberships_demo (
    member_id INT,
    member_name STRING,
    city STRING,
    membership_plan STRING,
    monthly_fee INT,
    duration_months INT,
    status STRING
)
USING DELTA;
```

Step 2 – Insert sample records

```
%sql
INSERT INTO fitness_memberships_demo VALUES
(101, 'Arjun', 'Hyderabad', 'Premium', 3000, 12, 'Active'),
(102, 'Sneha', 'Pune', 'Basic', 1000, 3, 'Active'),
(103, 'Kiran', 'Chennai', 'Standard', 2000, 6, 'Expired'),
(104, 'Meera', 'Hyderabad', 'Basic', 1000, 6, 'Active'),
(105, 'Vikram', 'Delhi', 'Premium', 3000, 12, 'Active'),
(106, 'Anjali', 'Pune', 'Standard', 2000, 3, 'Cancelled'),
(107, 'Rohit', 'Mumbai', 'Basic', 1000, 3, 'Active'),
(108, 'Divya', 'Chennai', 'Premium', 3000, 6, 'Expired'),
(109, 'Sanjay', 'Hyderabad', 'Standard', 2000, 12, 'Active'),
(110, 'Pooja', 'Delhi', 'Basic', 1000, 3, 'Cancelled'),
(111, 'Nikhil', 'Mumbai', 'Premium', 3000, 12, 'Active'),
```

```
(112, 'Kavya', 'Pune', 'Standard', 2000, 6, 'Active'),
(113, 'Manoj', 'Chennai', 'Basic', 1000, 3, 'Expired'),
(114, 'Riya', 'Hyderabad', 'Premium', 3000, 12, 'Active'),
(115, 'Deepak', 'Delhi', 'Standard', 2000, 6, 'Active'),
(116, 'Neha', 'Mumbai', 'Basic', 1000, 3, 'Cancelled'),
(117, 'Tarun', 'Pune', 'Premium', 3000, 12, 'Active'),
(118, 'Aisha', 'Chennai', 'Standard', 2000, 6, 'Active'),
(119, 'Harish', 'Hyderabad', 'Basic', 1000, 3, 'Expired'),
(120, 'Simran', 'Delhi', 'Premium', 3000, 12, 'Active');
```

Step 3 — Student exercises

Complete the following tasks using SQL in Azure Databricks. No solutions are provided.

Part A — Retrieving and filtering data

1. Display all records from the Delta table.
2. Retrieve only `member_name`, `city`, and `membership_plan`.
3. Display members from Hyderabad.
4. Find members whose monthly fee is greater than ₹2,000.
5. Display all distinct cities.
6. Find members belonging to the Premium or Standard plan using `IN`.
7. Retrieve members whose monthly fee is between ₹1,000 and ₹2,000.
8. Find members whose names start with `S` using `LIKE`.
9. Display Active members from Hyderabad using `AND`.
10. Find members from Pune or Chennai using `OR`.
11. Display member names as `Name` and monthly fees as `Fee` using column aliases.
12. Retrieve all Premium members using a table alias.

Part B — Modifying Delta table records

13. Insert a new member with ID 121, name Farhan, city Jaipur, Premium plan, monthly fee ₹3,000, duration 6 months, and Active status.
14. Update Sneha's monthly fee to ₹1,500.
15. Change Kiran's membership status from Expired to Active.
16. Increase the monthly fee of all Premium members by ₹500.
17. Delete all members whose status is Cancelled.
18. Delete the member whose ID is 113.

Part C — Delta table operations

19. Display the complete transaction history using `DESCRIBE HISTORY`.
20. Find the table version before the updates and retrieve its records using Time Travel.
21. Compare the current records with the earlier version.
22. Count the number of records remaining after all updates and deletions.